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8.0 Basic Administration

This section focuses on administrative or maintenance task of the Ubuntu server. In this laboratory hands-on, the root privilege is discussed. Root privileges allows the user to execute certain commands that can change the system. Use in conjunction with root privilege is the `sudo` command which allows overriding of system settings and assuming a user or root account. Root privileges also allow installing or updating applications in the system that uses the APT package manager. The APT package manager uses the `apt` command.

Running services can be started, stopped or check status using the `systemctl` command. Aside from managing running services, it can also change the runlevel of the system. Run levels are operational levels that describe the state of the system with respect to what services are available. This can be used to make the system run certain services for maintenance, or put in 'safe mode'.

Other basic administration tools used is `tar` command for archiving and backing-up of data, `cron` utility for automating scheduled task and `lshw` for checking browsing hardware devices on the system. Also, network configuration can be changed using the `netplan` service.

8.1. Root Privileges

Any Unix/Linux system has a root account which has command or system privileges. With the root account, you should be able to start-up or shut down your Unix/Linux server, add users and groups, set privileges to other users, add additional disks or manage the system. As a word of caution, work on your Unix/Linux with the least privileges account to make sure that you do not damage the system when you make a mistake.

In the Ubuntu Linux operating system, you can use the '`sudo`' command to execute commands that needs root privileges. The first user account that you create can use the '`sudo`' command.

Step 1 Login to your Ubuntu server installation and go to the "`/home`" directory. Inside the home directory, type in the command: "`mkdir data`".

1a. Was the directory created? Why or why not?

No. The permission was denied as we are not a sudoer.

Step 2 In the "`/home`" directory, type in the command: "`sudo mkdir data`". Enter your password when asked.

2a. Was the directory created? Why or why not?

Yes. The data directory was successfully created as we are now a sudoer.

Step 3 Exit your user account by using the "`exit`" command.

Aside from using the 'sudo' command in conjunction with another command to execute root privileges, there are other options that can be used with the command. The following are options with the 'sudo' command:

Table 8-1 Options for the sudo command

-b	-	runs a command in background
-i	-	spawns a root shell
-k	-	resets the timestamp of the user running 'sudo' commands, user will need to enter password next 'sudo' command is used
-s	-	spawns a root shell but does not change the environment

There are more options that can be used with the 'sudo' command, but these are the usual options to be used in any Ubuntu Linux system.

Step 4 Login to your account.

4a. On the prompt, type in the command 'whoami'. What was the output of the command? What is your user account name?

The output is 'administrator'. The user account name is administrator.

4b. On the prompt, type in the command 'pwd'. What was the output of the command? What is your current directory?

The output is our current directory, it is the /home/administrator directory.

4c. On the prompt, type in the command 'id'. What was the output of the command? What is your user id (uid), group id (gid) and the groups you belong to?

```
administrator@lbysyad-s18-20162:~$ id
uid=1000(administrator) gid=1000(administrator) groups=1000(administrator),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),110(lxd)
administrator@lbysyad-s18-20162:~$
```

The output is the uid, gid, and the groups that the user belongs to.
Uid = 1000
Gid = 1000
Groups = 1000 (administrator), 4 (adm), 24 (cdrom), 27 (sudo), 30 (dip), 46 (plugdev), 110 (lxd)

Step 5 On the prompt, type in the command 'sudo -s'. Enter your password when asked.

5a. On the prompt, type in the command 'whoami'. What was the output of the command? What is your user account name?

The output is the current user account name, which is 'root'.

5b. On the prompt, type in the command 'pwd'. What was the output of the command? What is your current directory? Did your directory change?

The output of the command is the current directory we are in, which is /home/administrator.

5c. On the prompt, type in the command 'id'. What was the output of the command? What is your user id (uid), group id (gid) and the groups you belong to?

```
root@lbysyad-s18-20162:/home/administrator# id
uid=0(root) gid=0(root) groups=0(root)
```

The output is the uid, gid, and the groups that the user belongs to

```
Uid = 0
Gid = 0
Groups = 0
```

5d. Type in the command '`exit`'.

Step 6 On the prompt, type in the command '`sudo -i`'.

6a. On the prompt, type in the command '`whoami`'. What was the output of the command? What is your user account name?

The output is the current account name, which is 'root'.

6b. On the prompt, type in the command '`pwd`'. What was the output of the command? What is your current directory? Did your directory change?

The output is the current directory, it changed to /root.

6c. On the prompt, type in the command '`id`'. What was the output of the command? What is your user id (`uid`), group id (`gid`) and the `groups` you belong to?

The output is the uid, gid, and the groups that the user belongs to
Uid = 0
Gid = 0
Groups = 0

6d. What is the difference between the commands '`sudo -s`' and '`sudo -i`'?

The sudo -i simulates the root shell, while sudo -s simulates it without changing the environment.

6e. Type in the command '`exit`'.

Step 7 On the prompt, type in the command '`sudo -k`'.

7a. On the prompt type in the command '`sudo adduser user02`' and give it the password same as the account name (`user02`). *(Note: For now, this command adds another user account. Also just press the 'enter' key when asked for details such as name, number, phone, and others to skip.)*. Select '`y`' when asked for confirmation.

7b. Were you asked to enter your password first? Why did the system ask for the password while it did not ask in the previous commands entered?

The system asked for the new password and to retype the new password. The system asked for a password as we are trying to add a new user in the prompt.

7c. On the prompt type in the command '`sudo adduser user03`' and give it the password same as the account name (`user03`). Did the system ask you to enter your password? Why or why not?

Yes, the system asked to enter the new password and to retype the new password because we are trying to create a new account.

7d. Exit your user account.

Not all user accounts can use the '`sudo`' command. By default, only the user account created during installation can use the command. By editing the `sudoers` file, other users can be given this privilege. In the `sudoers` file, the user privilege specification part allows you add users in the following format:

user_list host_list = [(runas_list)] command)_list

user_list - specifies the user or users this line applies to, this can be usernames, groups (must be specified with %) or user aliases

host_list	-	specifies the host or hosts this line applies to, this can be hostnames, ip addresses or host aliases
runas_list	-	specifies the user that the user in the *user_list* can run as, this can be usernames, groups (must be specified with %) or user aliases
command_list	-	specifies the commands that the user in the *user_list* can run, this can be utilities, names of directories holding utilities or command aliases Note: command format should be an absolute filename

Step 8 Login to your Ubuntu Linux system using the `user02` account.

8a. At the command prompt, type in the command '`cd /`'. Who is the owner of the '`home`' directory?

The owner is root.

8b. Enter the '/`home`' directory. At the command prompt, type in the command '`mkdir data1`'. Was your account allowed to create the directory? Why or why not?

User02 was not allowed to create the directory, because there are no given privileges.

8c. At the command prompt, type in the command '`sudo mkdir data1`'. Was your account allowed to use the '`sudo`' command? Why or why not?

```
user02@lbysyad-s18-20162:/home$ sudo mkdir data1
[sudo] password for user02:
user02 is not in the sudoers file.  This incident will be reported.
user02@lbysyad-s18-20162:/home$
```

The user was not allowed to create the directory with the sudo account as user02 is not in the sudoers file.

8d. Exit the user account user02.

Step 9 Login using your administrator user account and enter the command '`sudo visudo`'. This command allows you to add user accounts in the `sudoers` file.

9a. Under the section "*#User privilege specification*" line, add the following:

```
user02 ALL = (root)       ALL
user03 ALL = (root)       /bin/mkdir
```

Looking at the added lines, what are the commands the user02 and user03 can use and which user can they run the commands as?

User02 is given privileges to run all commands in the root, while user03 can only run commands in /bin/mkdir.

9b. Save the file then exit the user account.

Step 10 Login as `user02` into your Ubuntu Linux system.

10a. At the command prompt, type in the command '`cd /home`'. What is your current directory? Who is the owner of this directory?

The current directory is /home and the owner is root.

10b. At the command prompt, type in the command '`mkdir data1`'. Was your account allowed to create the directory? Why or why not?

User02 was not allowed to create data1 as permissions were denied.

10c. At the command prompt, type in the command '`sudo mkdir data1`'. Was your account allowed to use the '`sudo`' command? Why or why not?

User02 successfully created the data1 directory as they used their sudo privileges.

10d. At the command prompt, type in the command '`sudo rmdir data1`'. Was the directory removed? Why or why not?

The directory was successfully removed the data1 directory as they were using their sudo privileges.

10e. Exit the user account `user02`.

Step 11 Login as `user03` into your Ubuntu Linux system.

11a. At the command prompt, type in the command '`cd /home`'. What is your current directory? Who is the owner of the directory?

The current directory is /home, while the owner is root.

11b. At the command prompt, type in the command '`mkdir data1`'. Was your account allowed to create the directory? Why or why not?

User03 was not able to make the directory as permissions were denied when commands list were set.

11c. At the command prompt, type in the command '`sudo mkdir data1`'. Was your account allowed to use the '`sudo`' command? Why or why not?

User03 was successfully able to create the data1 directory as sudo mode is being used.

11d. At the command prompt, type in the command '`sudo rmdir data1`'. Was the directory removed? Why or why not? What is the difference between `user02` and `user03` accounts?

The directory was not removed as user03 is not allowed to execute such commands. The difference between these two users is that user02 has all permissions allowed, while user03 can only execute the mkdir command.

11e. Exit the user account `user03`.

8.2. APT Package Manager

The APT package manager allows search, installation, upgrade, and removal of packages / software. The following are APT commands that can be used:

```
apt <command> <argument>
```

where:

Command	Description
`install`	Installs a package from the Ubuntu repository
`remove`	Removes a package on the system

`purge`	Removes a package and its configuration
`update`	Refreshes the repository index of the Ubuntu repository
`upgrade`	Upgrades all the upgradable packages in the system
`search`	Searches for a package in the Ubuntu repository
`show`	Shows the details of a package

Step 12 Login to your default account on your Ubuntu Linux system. At the command prompt type in the command '`sudo apt update`'.

12a. What does the command do?

The command will prompt for the account password then initiate the sudo update.

12b. In the command prompt, type in the command '`gcc`'. Did the command start the GNU C compiler? Why or why not?

The command 'gcc' was not found as it is not installed.

12c. In the command prompt, type in the command '`apt search gcc`'. Include the command prefix '`sudo`' if the command was not accepted. Describe the output. What does the '`apt search`' command do? Did it allow you to search if a `gcc` package can be installed?

The command started enumerating numerous packages in apt that have 'gcc'. It allows the user to search if gcc can be installed.

12d. In the command prompt, type in the command '`sudo apt install gcc`

The command initiated the gcc installation after prompting the user if they want to continue.

12e. In the command prompt, type in the command '`gcc`'. Was the GNU C compiler installed?

The compiler was installed.

12f. How can the APT package manager be used in the Ubuntu server system?

It manages the software packages on the server by simplifying and updating it.

12g. Exit the default user account.

8.3. Runlevel, Services and Shutdown

Traditionally, most Linux system uses the System V init daemon (a daemon is a background process or program), the init daemon as a process ID (PID) of '1'. The init daemon spawns all other processes in the Linux system. The init daemon uses the concept of runlevels to start or stop services. Runlevels in the linux system can be from level 2 to 5. Each runlevel can have different services that are started, while runlevel 6 is halt or power off. Each runlevel configuration is stored in the '`/etc/rc?.d`' directory, where the ? symbol represents the runlevel number. In each runlevel directory are scripts that starts or stops services. Each script starting with a capital 'S' means start the service while a capital 'K' means stop or kill a service. The number beside the 'S' or 'K' symbol is the sequence for starting or stopping a service.

In the Ubuntu Linux system, the init daemon has been replaced with the 'systemd' program since the old init program is not able to handle plug and play or new hardware and software. systemd replaces has predefined groups of units known as **targets**. Targets are usually defined according to the intended use of the system, and ensure that required dependencies for that use are met.

Normally, the system boots to the target described in `/lib/systemd/system/default.target`. This file is a symbolic link that can be changed when booting to a different target is desired. In the systemd program, the runlevel can be changed by using the ‘`systemctl`‘ command. The following table shows some standard preconfigured targets:

Runlevel	**Target**
`0`	poweroff.target
`1`	rescue.target
`2, 3, 4`	multi-user.target
`5`	graphical.target
`6`	reboot.target

Runlevel Configuration and Commands

Step 13 Login to your system using the default account.

13a. At the command prompt type in the command '`sudo systemctl list-units --type target`'. How many '`units`' were loaded in the current target? List some of the units loaded.

There were 26 units loaded in the current target, some of the units loaded are basic, cloud-config, cloud-init, and swap.

13b. Enter the command '`runlevel`'. What is your current runlevel?

The current runlevel is N 5.

13c. At the command prompt, type in the command '`sudo systemctl isolate rescue.target`'. What happened to your Ubuntu Linux installation?

The Ubuntu Linux installation went blank and is now in rescue mode. The system prompted to enter for maintenance.

13d. Press 'Enter' to continue. Did the runlevel change? After a prompt has appeared, type in the command '`sudo systemctl list-units --type target`'. How many units were loaded? List down some of the units loaded

The runlevel changed to 5 1. There were 11 units loaded, some of the units loaded is cryptsetup, local-fs-pre, veritysetup, and rescue.

13e. Type in the command '`reboot`' and then relogin with your default account after booting.

13f. After logging in, check the current runlevel again. Did it go back to the same runlevel as in step 13b? Why?

Yes, it went back to N 5. This is the runlevel because we rebooted the system, N 5 is the runlevel when no changes have been made.

13g. At the command prompt, type in the command '`sudo sytemctl isolate reboot.target`'? What did the system do? Which runlevel target is this?

The system prompted for the password then it started rebooting the linux system. The runlevel target of that was 6 (reboot)

1a. Login to the system and power off the system using the '`systemctl`' command? What is the command?

The command is sudo systemctl poweroff.

8.3.1. Services

Login to your system using your default account. At the command prompt, type in the command '`sudo ifconfig`'. At the '`eth0`' or '`ens18`' line, determine your IP address.

Note: Linux may use a different device name for the LAN that the system is using. Ask the instructor which is the correct device name. If `ifconfig` is not working, you need to install it first by using the command `sudo apt install net-tools`

Step 2

13h. What is your IP address?

The IP address is 10.2.9.63 with subnet mask of 255.255.0.0

13i. At the command prompt, type in the command '`sudo systemctl status ssh`'. What is the status and PID of the ssh service?

The status is active and running, with PID of 135592.

13j. Use an ssh client like the 'PuTTY' software to connect to your Ubuntu Linux installation. Use the instructions given from the last activity. Was an ssh connection established to the system? Why or why not?

The SSH connection was successfully established, as the port was successfully redirected to 61063.

13k. Logout from your ssh connection.

13l. On your Ubuntu Linux system, type in the command '`sudo systemctl stop ssh`'. What was the output message?

There was no output but the command was entered successfully.

13m. At the command prompt, type in the command '`sudo systemctl status ssh`'. What is the status of ssh service?

The ssh service status is now inactive (dead).

13n. Connect to your Ubuntu Linux installation using an SSH client software such as PuTTY. Was an SSH connection established? Why or why not?

No, the connection was refused as the sudoer eliminated ssh services.

13o. Try to turn on the SSH service. What is the command used to enable SSH? Was a connection established after enabling SSH?

The command used to enable SSH is 'sudo systemctl start ssh'. The connection was then restored for SSH services.

8.3.2. Shutdown Command

Step 14 Login to your default account and type in the command '`sudo shutdown -h xx:yy`' where '`xx:yy`' is three minutes from now. **Note:** Your system might be using the UTC time zone. If this is the case, subtract 8 hours from the Philippine Standard Time. (Example: 12:45PM PST is 04:45AM UTC).

14a. Did the system shutdown after three minutes? Why or why not?

The system did in fact shutdown after three minutes, as the sudoer set the shutdown command at that time.

14b. Start your system and login to your default account. At the command prompt, type in the command '`sudo shutdown -h now`'. Did the system shutdown immediately?

Yes, the system initiated shutdown immediately after prompting for the password.

14c. Start your system and login to your default account. At the command prompt, type in the command '`sudo shutdown -r now`'. Did the system restart after shutting down? Why?

Yes, the system immediately started restarting because of the -r option.

14d. Start your system and login to your default account. At the command prompt, type in the command '`sudo reboot`'. Did the system restarting after shutting down? Why? What is a similar command?

Yes, the system started restarting as the command is similar to the 'sudo shutdown -r now' command.

14e. Start your system and login to your default account. At the command prompt, type in the command '`sudo poweroff`'. Did the system shutdown immediately? What is a similar command?

Yes, the system shutdown immediately after prompting for the password. The similar password is sudo shutdown -h now.

8.4. `timedatectl` Utility

The '`timedatectl`' utility allows you to check and modify the system date of your Ubuntu Linux system. You can also change your time zone using the utility. The '`timedatectl`' utility has a format of:

```
timedatectl <command> <argument>
```

where:

Command	Description
status	Shows the current date, time, time zone and other configuration on the system time.
list-timezones	List the available time zones that can be set on the system
set-timezone	Sets the time zone of the system
set-ntp	Turn on or off time synchronization on the system using NTP (Network Time Protocol)
set-time	Sets the time and date of the system.

Step 15 Login to your default account on your Ubuntu Linux system. At the command prompt, type in '`timedatectl status`'.

15a. What is your current system date and time zone?

The current date is Feb 20 05:18:31 in the UTC time zone.

15b. At the command prompt, type in the command '`timedatectl`'. Is this the same as the previous command? Why or why not?

Yes, the commands are similar. Status is the same as asking the prompt for the current time.

15c. At the command prompt, type in the command '`timedatectl list-timezones`'. What does the command do? Write down the appropriate time zone keyword for the Philippine Standard Time (+0800).

The command lists the possible timezones. The time zone keyword is Asia/Manila.

15d. At the command prompt, type in the command '`sudo timedatectl set-timezone <timezone>`'. Use the time zone that was written down in step 16c for '`<timezone>`'. Did the time zone change? Why or why not?

Yes, the time zone changed as we entered the command to set the different time zone.

15e. At the command prompt, type in the command '`sudo timedatectl set-time <hh:mm>`'. Enter a time that is five minutes ahead of the current time. Did the time on the system change? Why or why not?

The time did not change as Automatic time synchronization is enabled.

15f. At the command prompt, type in the command '`sudo timedatectl set-ntp off`'. What does this command do?

This command turns off the Automatic time synchronization.

15g. At the command prompt, type in the command '`sudo timedatectl set-time <hh:mm>`'. Enter a time that is five minutes ahead of the current time. Did the time on the system change? Why or why not?

Yes, the time on the system changed as we disabled the synchronization.

15h. At the command prompt, type in the command '`sudo timedatectl set-time <yyyy-mm-dd>`'. Enter a date that is one day ahead of the current date. Did the date on the system change? Why or why not?

Yes, the date on the system indeed changed because we set it one day ahead with the command.

15i. At the command prompt, type in the command '`sudo timedatectl set-ntp on`'. Did the system time change back to the correct date and time? Why or why not?

Yes, it changed back to the current time in our time zone as we turned synchronization back on.

8.5. `cron` Utility

The 'cron' utility allows an administrator or user to setup a program or application to be executed by the system at a specific time. The 'cron' utility can execute any program in the system and especially useful for creating back up of data or system files at specific time. The root account has automatic access to the 'cron' utility but users has to be given permission by the root.

The 'cron' utility configuration file is the '`/etc/crontab`' file. The administrator can edit the file by entering the date, time or day of the week and the program to be executed as specified by the time. Users can be given permission to enter programs or task to be executed in the '`/etc/cron.allow`' or '`/etc/cron.deny`'. If the '`/etc/cron.allow`' exists, users in the file are given permission to use the 'cron' utility while if the '`/etc/cron.deny`' exists, then users in the file are denied access to the 'cron' utility.

The '/etc/crontab' file can be edited using the command 'crontab –e'. An entry in the 'crontab' file looks like:

```
minute (0-59), hour (0-23, 0 = midnight), day (1-31), month (1-12), weekday (0-6, 0 = Sunday), command
```

An example of an entry can be:

```
01 04 1 1 1 /usr/bin/somedirectory/somecommand
```

This means the program 'somecommand' in the directory '/usr/bin/somedirectory' will be executed at 04:01am at January 1 plus every Monday in January. Asterisk can be used to substitute for the time, date or weekday. An example can be:

```
01 04 * * * /usr/bin/somedirectory/somecommand
```

This means that the program 'somecommand' will be executed at 04:01am at everyday of every month. Another option is to use a comma-separated value and a dash. As an example:

```
01,31 04,05 1-15 1,6 * /usr/bin/somedirectory/somecommand
```

This means that 'somecommand' will be executed at 01 minute and 31 minute of 04:00 am and 05:00am plus at dates 1 through 15 of every January and June. Using a slash (/) will run the command at a number of times per time unit. As an example:

```
*/10 * * * * /usr/bin/somedirectory/somecommand
```

This means that the command 'somecommand' will be executed every ten (10) minutes. Note that the command should always be specified using the absolute filename as much as possible. Other options of 'crontab' are:

Options	Description
`-l` -	Outputs the crontab file on the screen
`-r` -	Removes the current crontab file

Step 16 Login to your account and type in the command '`sudo crontab -l`'.

16a. What is the output? Does your system have any crontab entries?

The output is that there are no crontabs for root.

Step 17 At the command prompt enter the command '`sudo mkdir -p /home/out`'. Then change the permission using the command '`sudo chmod 777 /home/out`'.

17a. At the command prompt, type in the command '`sudo crontab -e`'. This will start the default editor program and allow you to edit the crontab file. Edit the file to have an entry:

```
mm hh * * * echo "Executed cron command at $(date)" >> /home/out/crontext.out
```

The '`mm hh`' should have a time that is three minutes from now. After saving the crontab file, restart the cron service using '`sudo systemctl restart cron.service`' and enter the command '`sudo crontab -l`'. What was the output? Was there an entry? How many entries are there?

The output was the whole /home/out file. There is one entry, the one we input in the file.

17b. After waiting for three minutes, check if the file `crontext.out` exists. If it exists, use the command '`cat /home/out/crontext.out`' to check the contents. What was the output? Is it the expected output?

The output showed the "Executed cron command". Yes, it is the expected output. The time stated is 3 minutes after

Step 18 Ensure that your time and date is correct before proceeding. To adjust the time and date of the system, use the '`timedatectl`' utility.

18a. Edit the crontab file using the command '`sudo crontab -e`'. Edit the file to have an entry:

```
00 13 DD MM * echo "Executed cron command at $(date)" >> /home/out/crontext.out
```

The '`DD MM`' should have a date one day from today. After saving the `crontab` file restart the `cron` service and enter the command '`sudo crontab -l`'. What was the output? How many entries are there?

The output was from the entry that we made to the file. There is one entry, which is from 18a

18b. Change the system time about 1 minute before the latest the entry and wait. Check if the `crontext.out` file exists and check its output. Was the output correct? What was the output?

The output was the crontext entry that we made to the file. The time stated in the output is not accurate, it

Step 19 Ensure that your time and date is correct before proceeding. To adjust the time and date of the system, use the '`timedatectl`' utility.

19a. Edit the `crontab` file using the command '`sudo crontab -e`'. Edit the file to have an entry:

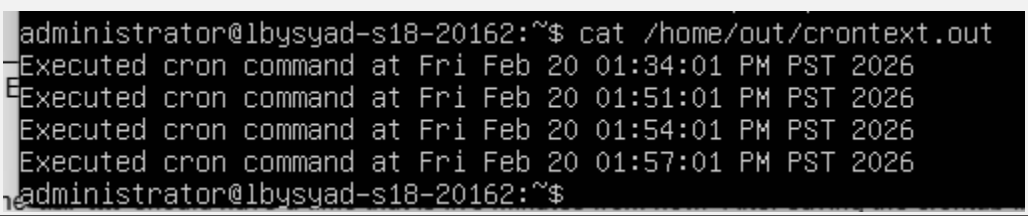
```
*/3 * * * * echo "Executed cron command at $(date)" >> /home/out/crontext.out
```

After saving the crontab file restart and enter the command '`sudo crontab -l`'. What was the output? How many entries are there?

The output was the content of the crontab file. There is one entry, it is the line we typed from 19a

19b. Wait for about 10 to 15 minutes and then check the output of the '`crontext.out`' file. Were there multiple new entries in the file? What was the time interval? Is it the expected output?

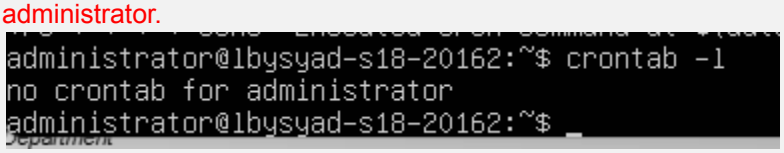
Yes, there are multiple entries in the file. The time interval is around 3-4 minutes per entry. This indeed the expected output.



Step 20 At the command prompt, type in the command '`sudo crontab -l`' and then type in the command '`crontab -l`'.

20a. What was the output? Were the outputs of the two commands same or different? Why?

The output for the sudoer is the content of the file, while the output without sudo has no crontab for administrator.



20b. Edit the `crontab` file (use '`crontab -e`' command) for your account and create an entry:

```
mm hh * * * echo "Executed cron command at $(date)" >> ~/crontext.out
```

The '`mm hh`' should have a time that is five minutes from now. After saving the crontab file, restart the `cron` service and enter the command '`crontab -l`' and '`sudo crontab -l`'. Were the outputs of the two commands same or different? Why?

The output of the two commands is the content of the file, however the time stamp outputs are different.

```
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
For more information see the manual pages of crontab(5) and cron(8)

m h  dom mon dow   command
*/3 * * * * echo "Executed cron command at $(date)" >> /home/out/crontext.out
```

```
administrator@lbysyad-s18-20162:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
06 14 * * * echo "Executed cron command at $(date)" >> ~/crontext.out
```

20c. Create a file in the '`etc`' directory and name it as '`cron.deny`'. Inside the file, put in your account name. Type in the command '`crontab -e`'. Are you able to edit your `cron` file? Why?

Yes, we were able to nano edit the file using the sudo command.

20d. Remove the '`cron.deny`' file.

Step 21 At the command prompt, edit the system `crontab` file and remove one of the entries. At the command prompt, type in the command '`sudo crontab -l`'.

21a. How many entries do you see?

There is one entry.

```
administrator@lbysyad-s18-20162:/etc$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
*/3 * * * * echo "Executed cron command at $(date)" >> /home/out/crontext.out
administrator@lbysyad-s18-20162:/etc$
```

21b. At the command prompt, type in the command '`sudo crontab -r`' and then type the command '`sudo crontab -l`'. Are there any entries?

The output is no crontab for root.

```
administrator@lbysyad-s18-20162:/etc$ sudo crontab -r
administrator@lbysyad-s18-20162:/etc$ sudo crontab -l
no crontab for root
administrator@lbysyad-s18-20162:/etc$ _
```

21c. At the command prompt, type in the command '`crontab -l`'. Are there any entries? Why?

No, the output is not being allowed to use this program as we put our account name in the cron.deny file

```
administrator@lbysyad-s18-20162:/etc$ crontab -l
You (administrator) are not allowed to use this program (crontab)
See crontab(1) for more information
administrator@lbysyad-s18-20162:/etc$
```

21d. At the command prompt, type in the command '`crontab -r`' and type in the command '`crontab -l`'. Are there any entries? Why?

There are no entries because we do not have the privileges to these commands. Hence, there were no entries

```
administrator@lbysyad-s18-20162:/etc$ crontab -r
You (administrator) are not allowed to use this program (crontab)
See crontab(1) for more information
administrator@lbysyad-s18-20162:/etc$ crontab -l
You (administrator) are not allowed to use this program (crontab)
See crontab(1) for more information
administrator@lbysyad-s18-20162:/etc$ _
```

21e. What is the difference between the '`crontab -r`' and '`sudo crontab -r`' commands?

The difference between the two commands is that sudo crontab -r opens the root's crontab and removes the file, while crontab -r simply says that you are not allowed to use this program. The command 'sudo' gives more privilege.

8.6. Back-Up and Restore (`tar` Utility)

The tar utility allows the user to pack files into one large file. The tar utility can also compress the file. The name tar comes from the term "tape archive". The tar utility is actually used to backup and restore files in the Unix/Linux system on a tape drive. The tar utility has several switches for options:

```
tar <-option> <argument 1> <argument 2>
```

where:

Options		Description
c	-	Create archive
v	-	Verbose
f	-	The target or source is a file
x	-	Extract file or files
z	-	Compressed

As an example, to archive the '/home/data' directory, use the command '`tar -vcf backup.tar /home/*`'. The command backs up the directory '`/home`' into '`backup.tar`'. The switch '`c`' creates the tar file, the switch '`f`' put the archive in a file (since traditionally the archive is put in a tape drive) while the '`v`' puts the command in verbose mode. Adding the '`z`' will compress the file. Normally, if you compress the archive, it should have a '`.tar.gz`' extension.

Step 22 Login to your account. On the command prompt, type in the command '`sudo tar -cvf backup1.tar /home/*`'. Take note that the '`sudo`' command is not really needed, it is used in this case to be able to access the other directories.

22a. Did the command creat the `backup1.tar` file? What is the file size?

Yes, the file size is 24G.

```
administrator@lbysyad-s18-20162:~$ df -h backup1.tar
Filesystem                         Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv   24G  8.2G   15G  37% /
administrator@lbysyad-s18-20162:~$ _
```

22b. At the command prompt, type in the command '`sudo tar -cvzf backup2.tar.gz /home/*`'. Did the command create the `backup2.tar.gz` file? What is the file size? Is it smaller or larger than the previous file?

```
administrator@lbysyad-s18-20162:~$ df -h backup2.tar.gz
Filesystem                         Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv   24G  8.2G   15G  37% /
administrator@lbysyad-s18-20162:~$ _
```

Step 23 Create an entry in the system crontab file that backs up the /home directory in format of '`backup-yyyy-mm-dd.tar.gz`' everyday at 5:00pm. The '`yyyy-mm-dd`' is the date today.

23a. What is the crontab entry you used?

The crontab entry used is 0 17 * * * backup-2026-02-23.tar.gz /home/

```
  GNU nano 6.2                                   /tmp/crontab.K2FNKw/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)

# m h  dom mon dow   command
0 17 * * * backup-2026-02-23.tar.gz /home/_
```

8.7. Network Configuration

Network configuration can be done in the Ubuntu Linux by editing a configuration file. The configuration file for networking is '`/etc/neplan`'. This file sets if the network device for DHCP or static IP. If static IP is to be used, the file contains the IP address, IP gateway and DNS IP address to be used. Before proceeding, take note of your current IP address (x.x.x.x) and subnet mask (/x notation). You can find these using the `ifconfig` command.

```
administrator@lbysyad-s18-20162:~$ cd /etc/netplan
administrator@lbysyad-s18-20162:/etc/netplan$ ifconfig
ens18: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.2.9.63  netmask 255.255.0.0  broadcast 10.2.255.255
        inet6 fe80::be24:11ff:fe38:fb11  prefixlen 64  scopeid 0x20<link>
        ether bc:24:11:38:fb:11  txqueuelen 1000  (Ethernet)
        RX packets 1767  bytes 422412 (422.4 KB)
        RX errors 0  dropped 1098  overruns 0  frame 0
        TX packets 191  bytes 28911 (28.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 104  bytes 8868 (8.8 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 104  bytes 8868 (8.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

administrator@lbysyad-s18-20162:/etc/netplan$
```

Step 24 Login to your account and the edit the `*.yaml` file inside '`/etc/netplan` to contain the following:

```
ethernets:
  ens18:
        dhcp4: false
        addresses:
```

```
      -  <assigned IP>/<prefix>
    routes:
      -  to: default
         via: <default gateway IP>
    nameservers:
      addresses: [<dns server IP>]
```

Note: Do not use tabs, only spaces. The brackets (`[ ]`) are included when typing addresses.

After editing the file, you will need to restart your system or restart the networking service by using the command '`sudo netplan apply`'.

24a. At the command prompt, type in the command '`sudo ifconfig`'. What does the command do? What is your IP address and netmask?

```
  GNU nano 6.2
# This is the network config written by 'subiquity'
network:
  ethernets:
    ens18:
      dhcp4: false
      addresses:
      - 10.2.9.63/16
      nameservers:
        addresses:
        - 8.8.8.8
        search: []
      routes:
      - to: default
        via: 10.2.0.1
  version: 2
```

```
administrator@lbysyad-s18-20162:/etc/netplan$ sudo netplan apply

** (generate:1239): WARNING **: 22:01:07.744: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible
 by others.
WARNING:root:Cannot call Open vSwitch: ovsdb-server.service is not running.

** (process:1237): WARNING **: 22:01:08.219: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible
 others.

** (process:1237): WARNING **: 22:01:08.412: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible
by others.

** (process:1237): WARNING **: 22:01:08.412: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible
by others.
administrator@lbysyad-s18-20162:/etc/netplan$ sudo ifconfig
ens18: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.2.9.63  netmask 255.255.0.0  broadcast 10.2.255.255
        inet6 fe80::be24:11ff:fe38:fb11  prefixlen 64  scopeid 0x20<link>
        ether bc:24:11:38:fb:11  txqueuelen 1000  (Ethernet)
        RX packets 2258  bytes 460820 (460.8 KB)
        RX errors 0  dropped 1424  overruns 0  frame 0
        TX packets 199  bytes 29523 (29.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 104  bytes 8868 (8.8 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 104  bytes 8868 (8.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

administrator@lbysyad-s18-20162:/etc/netplan$
```

The command shows our current ip address information, our ip address is currently 10.2.9.63 with a netmask of 255.255.0.0

24b. Find a partner group and try to ping each other using the private IP address. Were the pings successful?

No, because currently their IP address is not in the /etc/hosts file

Step 25 A static network hosts file can be used to keep track of IP addresses of other hosts for simple network configuration. On the '`/etc/hosts`' file, edit it to include the IP addresses of your partner group. Set their name to `partner`.

25a. After editing, try pinging your partner group's system using the name you assigned in the '`/etc/hosts`' file. Were you able to ping it? Why or why not?

Yes, we can ping group 11.

```
administrator@lbysyad-s18-20162:~$ ping 10.2.9.66
PING 10.2.9.66 (10.2.9.66) 56(84) bytes of data.
64 bytes from 10.2.9.66: icmp_seq=1 ttl=64 time=1.01 ms
64 bytes from 10.2.9.66: icmp_seq=2 ttl=64 time=0.485 ms
64 bytes from 10.2.9.66: icmp_seq=3 ttl=64 time=0.558 ms
64 bytes from 10.2.9.66: icmp_seq=4 ttl=64 time=0.517 ms
64 bytes from 10.2.9.66: icmp_seq=5 ttl=64 time=0.539 ms
```

Note: The hosts file is only good for simple network configuration, for complex networks, it will be better to use a DNS.

Step 26 The '`host.allow`' file allows the system administrator to tweak which hosts or network can access the service. Every entry in the file has the following format:

```
service : host/network : option [: option] ...
```

Where

service	-	Name of a program or daemon
host/network	-	Host IP or network address with netmask, fully qualified domain name or domain name
Option	-	Contains keywords like allow or deny

26a. Edit the '`/etc/hosts.allow`' file to allow SSH connection by creating an entry:

```
sshd: e.f.g.h : allow
```

The `e.f.g.h` is the IP address of your classmates' system or computer. Restart the network service (`sudo netplan apply`) before asking your classmate to try to connect. Was your classmate able to connect? Why or why not?

Yes, the partner group was able to establish an SSH connection using our assigned port number.

26b. Edit the '`hosts.allow`' file to deny your classmates SSH connection to your system. Do not forget to restart the network service. Was your partner able to connect? Why or why not?

Our partner group was not able to connect, because we denied them on hosts.deny .

There is equivalent deny file called '`hosts.deny`'. You can try to edit this file and see the result.

Step 27 Edit your network configuration back to DHCP configuration.

27a. What file or files did you edit? What does the file or files contain? What is your IP address, netmask, gateway, and DNS configuration?

We did not actually perform this as our instructor mentioned to use static addressing.

However here is the ideal answer, we need to edit the *.yaml file inside /etc/netplan tand set dhcp4 to true. Then, we would need to remove the static address.

```
  GNU nano 6.2
# This is the network config written by 'subiquity'
network:
  ethernets:
    ens18:
      dhcp4: false
      addresses:
      - 10.2.9.63/16
      nameservers:
        addresses:
        - 8.8.8.8
        search: []
      routes:
      - to: default
        via: 10.2.0.1
  version: 2
```

8.8. lshw Utility

The lshw utility list hardware information on your Ubuntu Linux. The name stands for “Hardware Lister”. The utility can show processor, memory, device and bus system configuration.

Step 28 Login to your account and type the command ‘`sudo apt install lshw`’. This command installs the ‘`lshw`’ utility. Do note that there is a possibility that it is already installed on the system.

28a. At the command prompt, type in the command ‘`sudo lshw -short`’. Describe the contents of the output.

The output shows the brief details on the hardware of the system.

```
administrator@lbysyad-s18-20162:~$ sudo lshw -short
H/W path              Device      Class      Description
=====================================================
                                  system     Standard PC (i440FX + PIIX, 1996)
/0                                bus        Motherboard
/0/0                              memory     96KiB BIOS
/0/400                            processor  QEMU Virtual CPU version 2.5+
/0/1000                           memory     4098MiB System Memory
/0/1000/0                         memory     4098MiB DIMM RAM
/0/100                            bridge     440FX - 82441FX PMC [Natoma]
/0/100/1                          bridge     82371SB PIIX3 ISA [Natoma/Triton II]
/0/100/1/0                        input      PnP device PNP0303
/0/100/1/1                        input      PnP device PNP0f13
/0/100/1/2                        storage    PnP device PNP0700
/0/100/1/3                        system     PnP device PNP0b00
/0/100/1.1            scsi1       storage    82371SB PIIX3 IDE [Natoma/Triton II]
/0/100/1.1/0.0.0      /dev/cdrom  disk       QEMU DVD-ROM
/0/100/1.2                        bus        82371SB PIIX3 USB [Natoma/Triton II]
/0/100/1.2/1          usb1        bus        UHCI Host Controller
/0/100/1.2/1/1        input5      input      QEMU QEMU USB Tablet
/0/100/1.3                        bridge     82371AB/EB/MB PIIX4 ACPI
/0/100/2              /dev/fb0    display    bochs-drmdrmfb
/0/100/3                          generic    Virtio memory balloon
/0/100/3/0                        generic    Virtual I/O device
/0/100/5                          bridge     QEMU PCI-PCI bridge
/0/100/5/1                        storage    Virtio SCSI
/0/100/5/1/0          scsi2       generic    Virtual I/O device
/0/100/5/1/0/0.0.0    /dev/sda    disk       53GB QEMU HARDDISK
/0/100/5/1/0/0.0.0/1  /dev/sda1   volume     1023KiB BIOS Boot partition
/0/100/5/1/0/0.0.0/2  /dev/sda2   volume     2GiB EXT4 volume
/0/100/5/1/0/0.0.0/3  /dev/sda3   volume     47GiB EFI partition
/0/100/12                         network    Virtio network device
/0/100/12/0           ens18       network    Ethernet interface
/0/100/1e                         bridge     QEMU PCI-PCI bridge
/0/100/1f                         bridge     QEMU PCI-PCI bridge
/1                    input0      input      Power Button
/2                    input1      input      AT Translated Set 2 keyboard
/3                    input3      input      VirtualPS/2 VMware VMMouse
/4                    input4      input      VirtualPS/2 VMware VMMouse
administrator@lbysyad-s18-20162:~$
```

28b. At the command prompt, type in the command '`sudo lshw -short > ~/hwinfo`'. What did the command do? What does the file '`hwinfo`' contain?

The command simply transferred the output of the 'sudo lshw -short' command into the 'hwinfo' file. Hwinfo contains a brief list of the hardware specs of our virtual machine.

28c. At the command prompt, type in the command '`sudo lshw -class network`'. What did the command do? Describe the contents of the output.

```
administrator@lbysyad-s18-20162:~$ sudo lshw -class network
  *-network
       description: Ethernet controller
       product: Virtio network device
       vendor: Red Hat, Inc.
       physical id: 12
       bus info: pci@0000:00:12.0
       version: 00
       width: 64 bits
       clock: 33MHz
       capabilities: msix bus_master cap_list rom
       configuration: driver=virtio-pci latency=0
       resources: irq:10 ioport:f060(size=32) memory:fea52000-fea52fff memory:fd604000-fd607fff memory:fea0000
     *-virtio1
          description: Ethernet interface
          physical id: 0
          bus info: virtio@1
          logical name: ens18
          serial: bc:24:11:38:fb:11
          capabilities: ethernet physical
          configuration: autonegotiation=off broadcast=yes driver=virtio_net driverversion=1.0.0 ip=10.2.9.63
```

The command shows the detailed network information of the system.

28d. At the command prompt, type in the command '`sudo lshw -class network > ~/nethwinfo`'. What did the command do? What does the file '`nethwinfo`' contain?

The command simply transferred the output of the 'sudo lshw -class network'' command into the nethwinfo file. nethwinfo contains specifications of the network hardware of the machine

```
administrator@lbysyad-s18-20162:~$ sudo lshw -class network > ~/nethwinfo
administrator@lbysyad-s18-20162:~$ cat nethwinfo
  *-network
       description: Ethernet controller
       product: Virtio network device
       vendor: Red Hat, Inc.
       physical id: 12
       bus info: pci@0000:00:12.0
       version: 00
       width: 64 bits
       clock: 33MHz
       capabilities: msix bus_master cap_list rom
       configuration: driver=virtio-pci latency=0
       resources: irq:10 ioport:f060(size=32) memory:fea52000-fea52fff memory:fd604000-fd607fff memory:fea00000-fea3ffff
     *-virtio1
          description: Ethernet interface
          physical id: 0
          bus info: virtio@1
          logical name: ens18
          serial: bc:24:11:38:fb:11
          capabilities: ethernet physical
          configuration: autonegotiation=off broadcast=yes driver=virtio_net driverversion=1.0.0 ip=10.2.9.63 link=yes multicast=yes
administrator@lbysyad-s18-20162:~$ sudo lshw -html > ~/hwinfo.html
Hardware Lister (lshw) -
```

28e. At the command prompt, type in the command '`sudo lshw -html > ~/hwinfo.html`'. What did the command do? What does the file '`hwinfo.html`' contain?

```
        <tr><td class="first">physical id: </td><td class="second"><div class="id">1</div></td></tr>
        <tr><td class="first">logical name: </td><td class="second"><div class="id">input0</div></td></tr>
        <tr><td class="first">logical name: </td><td class="second"><div class="id">/dev/input/event0</div></td></
        <tr><td class="first">capabilities: </td><td class="second"><dfn title="">platform</dfn> </td></tr>
 </tbody>    </table></div>
    </div>
<div class="indented">
        <div class="indented">
    <table width="100%" class="node" summary="attributes of input:1">
 <thead><tr><td class="first">id:</td><td class="second">input:1</td></tr></thead>
 <tbody>
        <tr><td class="first">product: </td><td class="second">AT Translated Set 2 keyboard</td></tr>
        <tr><td class="first">physical id: </td><td class="second">2</td></tr>
        <tr><td class="first">logical name: </td><td class="second">input1</td></tr>
        <tr><td class="first">logical name: </td><td class="second">/dev/input/event1</td></
        <tr><td class="first">logical name: </td><td class="second">input1::capslock</td></t
        <tr><td class="first">logical name: </td><td class="second">input1::numlock</td></tr
        <tr><td class="first">logical name: </td><td class="second">input1::scrolllock</td><
        <tr><td class="first">capabilities: </td><td class="second">i8042 PC AT keyboard controller">i
 </tbody>    </table></div>
    </div>
div class="indented">
        <div class="indented">
    <table width="100%" class="node" summary="attributes of input:2">
 <thead><tr><td class="first">id:</td><td class="second">input:2</td></tr></thead>
 <tbody>
        <tr><td class="first">product: </td><td class="second">VirtualPS/2 VMware VMMouse</td></tr>
        <tr><td class="first">physical id: </td><td class="second">3</td></tr>
        <tr><td class="first">logical name: </td><td class="second">input3</td></tr>
        <tr><td class="first">logical name: </td><td class="second">/dev/input/event3</td></
        <tr><td class="first">logical name: </td><td class="second">/dev/input/mouse1</td></
        <tr><td class="first">capabilities: </td><td class="second">i8042 PC AT keyboard controller">i
 </tbody>    </table></div>
    </div>
<div class="indented">
        <div class="indented">
    <table width="100%" class="node" summary="attributes of input:3">
 <thead><tr><td class="first">id:</td><td class="second">input:3</td></tr></thead>
 <tbody>
        <tr><td class="first">product: </td><td class="second">VirtualPS/2 VMware VMMouse</td></tr>
        <tr><td class="first">physical id: </td><td class="second">4</td></tr>
        <tr><td class="first">logical name: </td><td class="second">input4</td></tr>
        <tr><td class="first">logical name: </td><td class="second">/dev/input/event2</td></
        <tr><td class="first">logical name: </td><td class="second">/dev/input/mouse0</td></
        <tr><td class="first">capabilities: </td><td class="second">i8042 PC AT keyboard controller">i
 </tbody>    </table></div>
    </div>
</body>
</html>
administrator@lbysyad-s18-20162:~$ _
```

The command simply transferred the output of the ‘sudo lshw -html” command into the hwinfo.html file. Html contains specifications of hardware of the machine in a structured html webpage format, instead of a plaintext.

8.9. Conclusion/Reflection

Cite at least one advantage and one disadvantage of using `cron`.

The advantages of using cron include being able to automate tasks at a scheduled time (for accuracy) without needing to do any manual labor, while using little system resources. A disadvantage may be its lack of identifying errors, cron will not notify you if a job fails which may cause delays in the network.

What kind of tasks would you use `cron` for?

Cron is vital for repetitive and mundane tasks that can be scheduled like automatic server backups, restarting services at certain times, security logs, and other tasks that need to happen regularly.

Why is it important to set the correct date and time for your server?

It is important to set the correct date and time for the server as cron jobs rely on system time and date, and so that security logs would have an accurate time stamp. If the wrong time is set, the backups will be inconsistent, and you may not be able to track logs accurately.

Should all users be allowed to use `sudo`? Why or why not?

No, the sudo command gives elevated privileges to a user and enables them to modify sensitive files. Hence, it should only be used by limited and authentic users to ensure the security of the system. The sudo command should only be allowed to be used by trusted administrators